Yasin Torres-Tiji
Colorado School of Mines – Chemistry Department
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Professional Summary
Algal synthetic and systems biologist with end-to-end expertise spanning genomic analysis, gene design, transformation, bioreactor operation, and multi-omics characterization. Currently engineering *Picochlorum celeri* at the systems level as a postdoctoral scholar at Colorado School of Mines, with active collaborations across DOE national laboratories (PNNL, LANL, NLR, EMSL). PI on a DOE/EMSL campaign integrating transcriptomics, proteomics, and metabolomics to build predictive genome-scale metabolic models of algal productivity. Seeking faculty, PI, or staff scientist positions.

Education

University of California San Diego
Division of Biological Science
Ph.D. Molecular Biology - 2021

Autonomous University of Barcelona
B.S. Biotechnology – 2015

Additional Education

University of California San Diego
MicroMBA Program — 2023
Business analytics, corporate finance, marketing, operational strategy, and leadership.

National Center for Genome Resources
Differential Gene Expression Workshop — 2025
RNA-seq experimental design, read alignment, statistical analysis with DESeq2, and pathway enrichment.

Codecademy
Learn Python 3 — 2026
Programming fundamentals, data structures, control flow, functions, and file handling in Python.

Research Interests

Algal synthetic & systems biology · Genetic engineering · Multi-omics integration & genome-scale metabolic modeling · Bioreactor process engineering & climate simulation · Photosynthetic carbon metabolism & partitioning · Sustainable bioproduction & bioeconomy · AI/ML-assisted bioinformatics

Dissertation: "Optimizing recombinant protein production in *Chlamydomonas reinhardtii*: turning a model organism into a commercially relevant biotechnological host

Performed research in recombinant protein production in microalgae with expertise throughout the whole process: gene design, gene cloning, microalgal transformation with heterologous DNA, microalgal culturing, recombinant protein detection, recombinant protein purification and analysis. Focused on the optimization of two crucial elements of the process: improvement of recombinant protein production at single cell level by developing advanced genetic tools, and improvement of recombinant protein production by increasing the cell density in a heterotrophic bioreactor.

Skills

Molecular & Synthetic Biology

- End-to-end recombinant protein production: codon-optimized gene design, expression vector assembly, algal/bacterial transformation, high-expression clone selection, bioreactor scale-up, and protein purification.
- Gene design, molecular cloning, PCR, RT-qPCR, and nucleic acid extraction/purification.
- Genome editing (CRISPR), transgenesis, and cis-regulatory element identification in photosynthetic organisms.
- Western blot, chromatography, FPLC, ELISA, and mass spectrometry-based protein characterization.

Multi-Omics & Bioinformatics

- RNA-seq: experimental design, library preparation, STAR alignment, featureCounts, DESeq2 differential expression analysis, GO/KEGG pathway enrichment.
- TMT quantitative proteomics: LC-MS/MS data processing via FragPipe/MSFragger, limma-based differential expression analysis, TMT-Integrator for ratio quantification.
- Ubiquitination proteomics: site-level quantification and differential analysis (5,660+ sites), protein-abundance normalization, condition-specific detection workflows.
- Metabolomics: LC-MS (HILIC) and GC-MS data interpretation and integration.
- Genome annotation: multi-source pipelines integrating CLEAN, InterProScan, SwissProt BLAST, and eggNOG-mapper for functional characterization of non-model organisms.
- Genome-scale metabolic model construction and constraint-based analysis.
- Proficient in R and Python for bioinformatic analysis, statistical computing, and data visualization. Working environment: WSL2 Ubuntu with conda-managed pipelines (STAR, fastp, BLAST+, Diamond, samtools, bedtools).
- NGS data analysis, genome assembly, and variant calling using command-line tools and Galaxy.
- Flow cytometry experimental design and data analysis (FlowJo).

Analytical Chemistry

- FAME extraction and fatty acid profiling by GC-FID.
- ELISA-based protein quantification.
- Mass spectrometry data analysis and interpretation (LC-MS/MS, GC-MS).

Microscopy & Microbial Culturing

- Optical, fluorescence, and confocal microscopy.
- Microalgal cultivation across scales: agar plates, flasks, open pond systems, and photobioreactors.
- Microbial identification via 16S/ITS/18S sequencing, metagenomics, and whole-genome sequencing.
- Sterile technique and contamination-free culture maintenance.

Bioreactor Process Engineering

- Design, operation, and media optimization of batch and semi-continuous bioreactor systems.
- Mastery in operating climate-simulated photobioreactors for generating reproducible growth data under controlled light, temperature, and CO_2 regimes.
- Process optimization for biomass titer and recombinant protein yield.

Communication, Mentorship & Project Management

- PI on DOE/EMSL CAM campaign; experienced in grant writing, project planning, and execution.
- Mentored 15+ undergraduate students and 4 graduate students.
- Effective communicator of complex scientific concepts to technical and non-technical audiences.
- Peer reviewer for 9 journals including Microbial Cell Factories, Food Bioscience, PLOS ONE, and Frontiers in Plant Science.

Professional Experience

Postdoctoral Fellow - Posewitz Lab
Colorado School of Mines, Department of Chemistry, Golden, CO
July 2024 – Present

- Performing research in algal synthetic biology and algal bioprocess engineering to enhance the cost competitiveness of algal bioproducts.

Postdoctoral Scholar - Mayfield Lab
University of California San Diego, Division of Biological Sciences, La Jolla, CA
October 2021 – June 2024

- Performed research in synthetic biology to create genetic tools that perfect our capacity to genetically engineer microalgae for the improved production of biofuels, bioplastics precursors, food and therapeutics.
  Technical achievements: Successfully identified a series of DNA cis-regulatory elements using bioinformatics analysis, significantly enhancing transgene expression. When

integrated into a minimal promoter, these elements not only matched but surpassed the efficacy of current state-of-the-art genetic tools in synthetic promoter design.

PhD Student - Mayfield Lab
University of California San Diego, Division of Biological Sciences, La Jolla, CA
September 2015 – September 2021

- Performed research developing genetic tools to improve transgene expression in the nuclear genome of Chlamydomonas reinhardtii and advanced the field of recombinant protein production in bioreactors using algae.
  Technical achievements: 2-fold improvement on biomass yield in a bioreactor, 3-fold improvement on recombinant protein titers compared to highest reported in the literature. Novel genetic tool producing a 10-fold improvement in transgene expression compared to state of the art.

Research Assistant - Mayfield Lab
University of California San Diego, Division of Biological Sciences, La Jolla, CA
January 2015 – June 2015

- Performed research developing genetic tools for recombinant protein production in the algal chloroplast of *Chlamydomonas reinhardtii.*

Research Assistant
Center for Research in Agricultural Genomics (CRAG), Barcelona, Spain
August 2012 – May 2014

- Performed research revolving the genetic regulation of lignin formation in corn to make second generation biofuels more economically viable.

Funding & Awards

U.S. Department of Energy (DOE) - Environmental Molecular Sciences Laboratory (EMSL) Grant No. 62132 (Feb 2026–Present)
Project: Genome-Scale Modeling of Picochlorum celeri to Elucidate CO_2-Dependent Extreme Growth and Carbon Partitioning in Algae
Richland, WA, USA. DOI: 10.46936/camc.proj.2026.62132/60015765
Role: Principal Investigator

Grant Banco Santander Formula 2014/2015 (5000 EUR award for studying abroad - 1 out of 3 recipients for the Autonomous University of Barcelona)

University of California Education Abroad Program 2014/2015 (1 out of 25 recipients for the Autonomous University of Barcelona)

Frontiers of Innovation Scholars Program 2016/2017 (25,000 USD fellowship University of California San Diego)

Publications

Torres-Tiji, Y., Fields, F. J., & Mayfield, S. P. (2020). Microalgae as a future food source. Biotechnology Advances, 41, 107536. https://doi.org/10.1016/j.biotechadv.2020.107536 **Citations: 607**

Diaz, C. J., Douglas, K. J., Kang, K., Kolarik, A. L., Malinovski, R., Torres-Tiji, Y., Fields, F. J., & Mayfield, S. P. (2023). Developing algae as a sustainable food source. Frontiers in Nutrition, 9, 3147. https://doi.org/10.3389/fnut.2022.1029841 **Citations: 175**

Berndt, A. J., Smalley, T. N., Ren, B., Simkovsky, R., Badary, A., Sproles, A. E., Torres-Tiji, Y., Melis, A., & Mayfield, S. P. (2021). Recombinant production of a functional SARS-CoV-2 spike receptor binding domain in the green algae Chlamydomonas reinhardtii. PloS One, 16(11), e0257089. https://doi.org/10.1371/journal.pone.0257089 **Citations: 50**

Torres-Tiji, Y., Fields, F. J., Yang, Y., Heredia, V., Horn, S. J., Keremane, S. R., Hannon, M. J., & Mayfield, S. P. (2022). Optimized production of a bioactive human recombinant protein from the microalgae Chlamydomonas reinhardtii grown at high density in a fed-batch bioreactor. Algal Research, 66, 102786. https://doi.org/10.1016/j.algal.2022.102786 **Citations: 35**

Gupta, A., Kang, K., Pathania, R., Saxton, L., Saucedo, B., Malik, A., Torres-Tiji, Y., Diaz, C. J., Dutra Molino, J. V., & Mayfield, S. P. (2024). Harnessing genetic engineering to drive economic bioproduct production in algae. Frontiers in Bioengineering and Biotechnology, 12, 1350722. https://doi.org/10.3389/fbioe.2024.1350722 **Citations: 41**

Torres-Tiji, Y., Sethuram, H., Gupta, A., McCauley, J., Dutra-Molino, J.V., Pathania, R., Saxton, L., Kang, K., Hillson, N. J., & Mayfield. S. P. (2024) Bioinformatic prediction and high throughput in vivo screening to identify cis-regulatory elements for the development of algal synthetic promoters. ACS Synthetic Biology, 13(7), 2150–2165. https://doi.org/10.1021/acssynbio.4c00199 **Citations: 9**

Molino, J. V. D., Oliver, A., Sethuram, H., Kang, K., Saucedo, B., Diaz, C. J., Gupta, A., Jen, L. J., Torres-Tiji, Y., Hidasi, N., Badary, A., Jenkins, H., Fields, F. J., Simkovsky, R., & Mayfield, S. P. (2025) Description of a novel extremophile green algae, Chlamydomonas pacifica, and its potential as a biotechnology host. Algal Research, 88, 104034. https://doi.org/10.1016/j.algal.2025.104034 **Citations: 6**

Gupta, A., Oliver, A., Dutra Molino, J. V., Wnuk-Fink, K. M. J., Tessman, M., Kang, K., do Espirito Santo, É., Torres-Tiji, Y., Burkart, M. D., & Mayfield, S. P. (2025) Identification and overexpression of endogenous transcription factors to enhance lipid accumulation in the biotechnologically relevant species Chlamydomonas pacifica. Algal Research, 90, 104192. https://doi.org/10.1016/j.algal.2025.104192 **Citations: 0**

Torres-Tiji, Y., Fields, F. J., & Mayfield, S. P. (2025) Microalgal Colony Blot – A Simple and Rapid Method for Direct Detection of Recombinant Proteins in Microalgae. PLOS One – Lab Protocols https://dx.doi.org/10.17504/protocols.io.kxygx9k2kg8j/v1 **Citations: 0**

Doctoral Thesis

Torres-Tiji, Y. (2021). Optimizing Recombinant Protein Production in Chlamydomonas reinhardtii: Turning a Model Organism into a Commercially Relevant Biotechnological Host. Doctoral Thesis, University of California, San Diego.

Other works

Torres Tiji, Y. (2015). Viability analysis of algae biofuels and assessment of the future energy situation.

Conferences/Public Talks

U.S. Department of Energy – Algae Productivity Exceeding Expectations Challenge Meeting - 2023

Algal Biomass, Biofuels and Bioproducts – 2023 - Poster

6th International Congress Meredith Gould – 2023 - Presentation

PepTalk – The Protein Science and Production Week – 2022 – Presentation

Algae Biomass Organization Summit 2020 – Presentation

Frontiers of Innovations Scholars Program – 2017 - Presentation

Peer Review Contributions

Microbial Cell Factories (Springer Nature) – 1 review (2026)

Food Bioscience (Elsevier) — 1 review (2025)

Journal of Agricultural and Food Chemistry (American Chemical Society) — 1 review (2025)

Journal of Biotechnology (Elsevier) — 1 review (2025)

Limnology and Oceanography: Methods (ASLO) — 1 review (2024)

PLOS ONE — 2 reviews (2024)

Frontiers in Genome Editing (Genome Editing in Plants) — 1 review (2025)

Frontiers in Plant Science (Plant Biotechnology) — 1 review (2025)

Critical Reviews in Biotechnology — 1 review (2020)

Mentorships

ENLACE Mentorship Program - Summer 2019

ENLACE Mentorship Program – Summer 2023

Mentored over 15 undergraduate students, and 4 graduate students.

Contact References for Yasin Torres-Tiji

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